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System and method for power management in solid state storage systems

Abstract

Systems and apparatus are provided for power management of non-volatile storage (NVM) systems. A storage controller includes a command queue and a processor. The processor is configured to, in response to a data transfer rate from a host falling below a predetermined threshold and the command queue containing no command, enter a pseudo-idle power state by turning off input and output (IO) drivers of a memory interface coupling the storage controller and a non-volatile memory device and keeping on an internal memory of the storage controller.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2006/0209444	12/2005	Song	N/A	G11B 20/10527
2007/0079063	12/2006	Mizuno	711/170	G06F 3/067
2008/0168223	12/2007	Reeves	711/E12.001	G06F 3/0674
2017/0068480	12/2016	Huang	N/A	G06F 1/3275
2017/0115916	12/2016	Jayachandran	N/A	G06F 3/0679
2022/0291843	12/2021	Kachare	N/A	G06F 3/0604

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Background/Summary

TECHNICAL FIELD

(1) The disclosure herein relates to management of non-volatile storage systems, particularly relates to power management for non-volatile storage systems.

BACKGROUND

(2) Fast, reliable non-volatile memory (NVM) based storage systems, such as solid state drives (SSDs), have revolutionized data storage. More recently, NVM Express (NVMe) is an industry standard that provides a super-fast way to access non-volatile memory. However, there's a downside to NVMe technology: high power consumption. Therefore, non-volatile storage systems have to implement power management features that find acceptable balances between power, performance, product reliability and customer experience.

(3) In general, a non-volatile storage system may be operational or idle, and thus, various operational and idle power levels have been defined. When a non-volatile storage system is non-operational but fails to transition to lower power modes, it will continue to draw power even though the non-volatile storage system is idle. On the other hand, while idle power states provide various power-saving levels, there's a tradeoff in the time it takes to enter and exit each idle power state. Long entry and exit latencies lead to lower performance or longer response time. Therefore, there is a need in power management for non-volatile storage drives that provides a balanced approach for power saving and entry/exit latencies.

SUMMARY

(4) The present disclosure provides a storage system that implements a pseudo-idle power state.

This pseudo-idle power state may be positioned as an intermediate power state between active and idle power states, with an entry latency and an exit latency at least one order faster than any idle power state. In the pseudo-idle power state, the power consumption may be about one order (5.sup.~20 times) lower than the lowest power consumption of any active state and one order higher than the highest power consumption of any idle state.

(5) In various embodiments, methods and systems may be implemented to manage the entry into and exit from the pseudo-idle power state. A threshold data transfer rate for the data bus connected to a host may be pre-determined as one condition to enter the pseudo-idle power state. A non-empty command queue for command received from the host may be used as another condition to enter the pseudo-idle power state. The storage system may enter this pseudo-idle power state when the command queue is empty, and the bus transfer rate is below a predetermined threshold but above zero. The command queue may also be used as the condition to exit the pseudo-idle power state and enter an active power state. When there is a command from the host, an exemplary non-volatile storage system may exit the pseudo-idle power state and enter a full performance power state.

(6) In one exemplary embodiment, there is provided a non-volatile storage system that may comprise: a first interface to be coupled to a host, a non-volatile memory device, a storage controller comprising a command queue and a processor, and a second interface coupling the storage controller and the non-volatile memory device. The processor may be configured to: handle data transfer requests from the host in an active power state, monitor the command queue and a data transfer rate on a data bus of the first interface, determine that the data transfer rate falls below a predetermined threshold and there is no command in the command queue, enter a pseudo-idle power state in which input and output (IO) drivers of the second interface is turned off but a power domain for an on-chip internal memory of the storage controller is kept on, determine that there is a new command in the command queue from the host, and exit the pseudo-idle power state and enter the active power state.

(7) In another exemplary embodiment, there is provided a method for managing power of a non-volatile storage system. The method may comprise handling data transfer requests from a host in an active power state, monitoring a command queue in a storage controller of the non-volatile storage system and a data transfer rate on a data bus of a first interface between the host and the non-volatile storage system, determining that the data transfer rate falls below a predetermined threshold and there is no command in the command queue, entering a pseudo-idle power state in which input and output (IO) drivers of a second interface coupling the storage controller and a non-volatile storage device of the non-volatile storage system is turned off but a power domain for an on-chip internal memory of the storage controller is kept on, determining that there is a new command in the command queue from the host, and exiting the pseudo-idle power state and entering the active power state.

(8) In yet another exemplary embodiment, there is provided a non-transitory machine-readable medium having computer instructions embedded thereon. The computer instructions, when executed by a hardware processor, may cause the hardware processor to perform: handling data transfer requests from a host at a non-volatile storage system in an active power state, monitoring a command queue in a storage controller of the non-volatile storage system and a data transfer rate on a data bus of a first interface between the host and the non-volatile storage system, determining that the data transfer rate falls below a predetermined threshold and there is no command in the command queue, entering a pseudo-idle power state in which input and output (IO) drivers of a second interface coupling the storage controller and a non-volatile storage device of the non-volatile storage system is turned off but a power domain for an on-chip internal memory of the storage controller is kept on, determining that there is a new command in the command queue from the host, and exiting the pseudo-idle power state and entering the active power state.

Description

BRIEF DESCRIPTION OF FIGURES

- (1) FIG. 1 schematically shows a non-volatile storage system with power management in accordance with an embodiment of the present disclosure.
- (2) FIG. 2 is block diagram of power states of a non-volatile storage system in accordance with an embodiment of the present disclosure.
- (3) FIG. 3 is a flowchart of a process for conducting power management of a non-volatile storage system in accordance with an embodiment of the present disclosure.

DETAILED DESCRIPTION

- (4) Specific embodiments according to the present disclosure will now be described in detail with reference to the accompanying figures. Like elements in the various figures are denoted by like reference numerals for consistency.
- (5) FIG. 1 schematically shows a non-volatile storage system **100** with power management including a pseudo-idle state in accordance with an embodiment of the present disclosure. The non-volatile storage system **100** may comprise a storage controller **102** and one or more non-volatile memory (NVM) devices **104**. The storage controller **102** may be coupled to the one or more NVM devices **104** by a NVM device interface **116**. The NVM device interface **116** may include input and output drivers (IO) and on-die termination (ODT) circuits. In embodiments the NVM devices are NAND devices, the NVM device interface **116** may be a NAND interface.
- (6) The non-volatile storage system **100** may be coupled to a host **106** via an interface **108**. The interface **108** may include a data bus for data transfer between the non-volatile storage system **100** and the host **106**. The interface **108** may be one of many currently used or yet to be developed interfaces, such as, but not limited to, Serial AT Attachment (SATA), Parallel ATA (PATA), PCI Express (PCIe), Secure Digital (SD), Universal Serial Bus (USB), etc. The non-volatile storage system **100** may be implemented in memory cards, USB flash drives and solid-state drives in various embodiments to provide non-volatile storage for the host **106**. In some embodiments, the host **106** may be a computing device, such as, but not limited to, a smart phone, a pad computing device, a desktop computer, a laptop computer, a server computer, etc.
- (7) The NVM devices **104** may provide non-volatile data storage for the non-volatile storage system **100**. In some embodiments, the NVM devices **104** may be one or more NAND flash devices. In some other embodiments, the NVM devices **104** may be one or more other types of non-volatile storage devices, for example, NOR flash memories, magnetoresistive random Access Memory (MRAM), resistive random access memory (RRAM), phase change random access memory (PCRAM), Nano-RAM, etc.
- (8) The storage controller **102** may comprise a computer processor **110**, a memory **112** and a command queue **114**. The computer processor **110** may be, for example, a microprocessor or a microcontroller. The memory **112** may be an on-chip Static Random Access Memory (SRAM) and may store executable instructions and data of a firmware that manage the non-volatile storage system **100**. In at least one embodiment, the computer processor **110** may be configured to execute executable instructions of the firmware in the memory **112** to perform various operations for managing the non-volatile storage system **100**. The command queue **114** may be a storage for commands from the host **104**. In the embodiments that the interface **108** is the PCI Express (PCIe), all commands from the host may be referred to as Non-Volatile Memory Express (NVMe) commands. The NVMe commands may include various sets of commands, for example, admin commands and I/O commands. The admin commands may include administrative and management commands, including identify and get-log-page, etc. The I/O commands may handle data access such as read and write.
- (9) The storage controller **102** may be configured to perform power management for the non-

volatile storage system **100**. In some embodiments, the non-volatile storage system **100** may have one or more active power states. In each of the one or more active power states, the non-volatile storage system **100** may be fully operational and actively handling data requests from the host **104**. For example, data requests may be received from the host **104** as data access commands. The storage controller **102** may pick up the commands from the command queue **114** and service the data requests (e.g., read data from or write data to the NVM devices **104**).

(10) In some embodiments, the one or more active power states may include Power State Zero (PS0), Power State One (PS1) and Power State 2 (PS2). PS0 may be an active state with full speed and performance and the highest power consumption. When the non-volatile storage system **100** is in PS0 power state, various components of the non-volatile storage system **100** may be operational, for example, the memory **112**, NVM device IO and ODT circuits of the NVM device interface **116**, various power domains of the controller **102** may be turned on. Moreover, there is no clock gating for the controller clock and the clock speed of the controller **102** may be high. In contrast, PS1 and PS2 may have some levels of throttling to limit power consumption as well as performance. For example, the clock speed of the controller **102** may be slower in PS1 than in PS0, and slower in PS2 than in PS1.

(11) The non-volatile storage system **100** may also have one or more idle power states. For example, the one or more idle power states may include Power State 3 (PS3), and Power State 4 (PS4). PS3 may be referred to as a sleep state, in which the non-volatile storage system **100** may enter a sleep mode. For example, the memory **112** may be put in a sleep (with power turned off but periodically turned on to refresh), the NVM device IO and ODT may be turned off, various power domains of the controller **102** may be partially on and partially off.

(12) Moreover, clock gating may be enabled for the controller clock and the clock speed of the controller **102** may be extremely low. PS4 may be referred to as a deep sleep state, in which in addition to those turned off in the PS3 state, more components of the non-volatile storage system **100** may be turned off. For example, various power domains of the controller **102** may be turned off.

(13) In addition to the one or more active power states and one or more idle power states, the non-volatile storage system **100** may have at least one power state which may consume more power than any idle power state and less power than any active power state. The power state with power consumption between the active power states and the idle power states is neither an active power state, nor an idle power state, and may be referred to as a pseudo-idle power state. In the pseudo-idle power state, more components of the non-volatile storage system **100** may be turned on compared to the idle power states, and less components of the non-volatile storage system **100** may be turned on compared to the active power states. For example, in the pseudo-idle power state, the memory **112** may be on, the NVM device IO and ODT may be turned off, various power domains of the controller **102** may be on. Moreover, in the pseudo-idle power state, clock gating may be enabled for the controller clock and the clock speed of the controller **102** may be low.

(14) Table 1 may illustrate some exemplary features of the pseudo-idle power state, PS0 and PS3 and show that the pseudo-idle power state may be distinctive from the existing idle states. First, the internal memory **112** (SRAM) of the controller **102** may stay on during the pseudo-idle power state. In contrast, the internal SRAM in the PS3 power state may be put into a deep sleep mode. Second, the NVM device IO drivers and on-die termination (ODT) may be switched to off for power savings in the pseudo-idle power state and may remain off in the idle power states.

(15) Third, the power domains inside the controller **102** may remain on in the pseudo-idle power state, while in PS3 some of the power domains may be turned off. For example, the power domain of memory controller is turned off in PS3, but may be kept on in the pseudo-idle power state.

(16) Forth, the controller clock gating may be enabled in the pseudo-idle power state for further power savings, while in the active states they may be disabled. Therefore, the pseudo-idle power state may have a shorter exit latency (e.g., shorter time to exit) than the idle state PS3. Because PS3

may have a shorter exit latency than PS4, the pseudo-idle power state may have a much shorter exit latency than the idle state PS4. That is, the non-volatile storage system **100** may exit from the pseudo-idle power state and enter PS0 (or any other active power states) faster than PS3 and PS4.

(17) TABLE-US-00001 TABLE 1 Implementation Details of the Pseudo-Idle Power State in one embodiment

Power States	PS3	Pseudo-Idle	Power State	PS0	SRAM	Deep Sleep	On	On	NVM
device IO	Off	Off	On	Controller	Power	Partially	Off	On	On
Domains	Controller	Clock	Enabled	Enabled	Off	Gating	Clock	Extremely	Low
	Low	Low	High						

(18) Also as illustrated in Table 1, the internal clock speed for the pseudo-idle power state may be lower than the internal clock speed for active states and higher than the internal clock speed for PS3. In one exemplary embodiment, the internal CPU clock for the controller **102** may be 166 MHz for the pseudo-idle power state, compared to 400.sup.~800 MHZ in the active states and only several MHz in the PS3 idle state.

(19) Data transfer via the data bus of the interface **108** may be monitored to determine whether the non-volatile storage system **100** should enter the pseudo-idle power state from an active power state. For example, the non-volatile storage system **100** may switch from an active power state when the data transfer rate on the interface **108** falls below a predetermined threshold value but still higher than zero and there is no command in the command queue **114**. In one embodiment, the data bus of the interface **108** may be monitored by the processor **110** (e.g., executing firmware). The processor **110** may calculate an average bus transfer rate by calculating the size of data transferred within a given time interval (e.g., one second). In some embodiments, the average bus transfer rate may be used to monitor the data bus and the predetermined threshold value may be 500 mega-bytes per second (MB/s). The data transfer may be monitored by checking the bus transfer rate at a frequency of once every millisecond (e.g., 1 KHz). If a new command is received from the host **106** and entered in the command queue **114**, regardless of whether it is an administrative command or an I/O command, the non-volatile storage system **100** may exit the pseudo-idle power state and enter an active state.

(20) FIG. 2 is block diagram of power states of a non-volatile storage system in accordance with an embodiment of the present disclosure. Power State zero (PS0) may be an active power state with full speed and full performance. The non-volatile storage system **100** may switch from PS0 to PS3 when the idle time of the interface **108** has reached a first threshold. The interface **108** may be idle when its data transfer rate is zero. In some embodiments, the first threshold may be in the order of seconds, for example, in a range of 1 to 3 seconds (1.sup.~3 seconds).

(21) The non-volatile storage system **100** may switch from PS3 to PS4 when the idle time of the interface **108** has reached a second threshold. In some embodiments, the second threshold may be in the order of tens of seconds, for example, in a range to 10 to 30 seconds (10.sup.~30 seconds).

(22) In various embodiments, the non-volatile storage system **100** may switch from PS0 to the pseudo-idle power state when the data transfer rate of the interface **108** falls below a predetermined threshold (e.g., 500 MB/s) and there is no command in the command queue **114**. If there is a new command from the host **106**, the non-volatile storage system **100** may exit the pseudo-idle power state and return to the active PS0 power state.

(23) The non-volatile storage system **100** may exit PS3 power state and PS4 power state and return to the active PS0 power state when data transfer has resumed on data bus of the interface **108** (e.g., after a new data access command has been received and processed). The non-volatile storage system **100** may also switch from the pseudo-idle state to the PS3 power state when the data transfer rate on the data bus of the interface **108** has reached zero and stayed idle for a threshold value (e.g., 1.sup.~3 seconds).

(24) The non-volatile storage system **100** may have other active power states, and the power state PS0 may be a representative power state for these active power states. The conditions for transitions (e.g., to and from) between any of the other active power states and pseudo-idle power state may be the same as PS0 and pseudo-idle power state.

(25) FIG. 3 is a flowchart of a process **300** for conducting power management of a non-volatile storage system in accordance with an embodiment of the present disclosure. At block **302**, data requests from a host may be handled by a non-volatile storage system in an active power state. For example, the non-volatile storage system **100** may handle data requests from the host **106** in an active power state. The data requests may be I/O commands received from the host to write data to or read data from the non-volatile storage system **100**.

(26) At block **304**, a data transfer rate on a data bus connected to the host and a command queue for commands from the host may be monitored. For example, the interface **108** may have a data bus coupled to the host and the data transfer rate on the data bus of the interface **108** may be checked periodically (e.g., by calculating the size of data transferred every second). Also, the command queue **114** may be monitored by the controller **102**. The data transfer may be monitored by checking the bus transfer rate at a frequency of once every millisecond (e.g., 1 KHz). At block **306**, it may be determined that the data transfer rate falls below a predetermined threshold value and there is no command in the command queue. For example, the controller **102** may determine that there is no command in the command queue, and the data transfer rate has fallen below a predetermined threshold (e.g., 500 MB/s).

(27) At block **308**, the non-volatile storage system may enter a pseudo-idle power state in which input and output (IO) drivers of a NVM device interface may be turned off but a power domain for an on-chip internal memory of the storage controller may be kept on. In various embodiments, the non-volatile storage system **100** may use the data transfer rate on the data bus of the interface connected to the host and whether there is a command in the command queue **114** as criteria to determine whether to enter the pseudo-idle power state. As soon as the data transfer rate falls below a predetermined threshold value and there is no command in the command queue **114**, the non-volatile storage system **100** may enter the pseudo-idle power state, in which the non-volatile storage system **100** may turn off certain functions to become not fully operational but not idle either, for example, I/O drivers for NVM device interface turned off but the power domain for the SRAM of the storage controller may be kept on.

(28) At block **310**, it may be determined that there is a new command from the host. At block **312**, the non-volatile storage system may exit the pseudo-idle power state to enter the active power state. For example, the non-volatile storage system **100** may monitor the command queue **114** and as soon as a new command is received from the host **106** and entered into the command queue, the non-volatile storage system **100** may leave the pseudo-idle power state and enter the active power state.

(29) The process **300** and features of the non-volatile storage system **100** related to power management (e.g., transition between different power states) may be implemented using software (e.g., executable by a computer processor (CPU, GPU, or both)), hardware (e.g., a field-programmable gate array (FPGA) or an application-specific IC (ASIC), firmware, or any suitable combination of the three. In one embodiment, for example, the process **300** and features of the non-volatile storage system **100** related to power management may be programmed in computer processor executable instructions, stored in a non-transitory machine-readable medium (e.g., the non-volatile storage device **104**, hard drive, CD, DVD, etc.) and performed by the processor **110** (e.g., a microprocessor or a microcontroller) executing the executable instructions.

(30) In one exemplary embodiment, there is provided a non-volatile storage system that may comprise: a first interface to be coupled to a host, a non-volatile memory device, a storage controller comprising a command queue and a processor, and a second interface coupling the storage controller and the non-volatile memory device. The processor may be configured to: handle data transfer requests from the host in an active power state, monitor the command queue and a data transfer rate on a data bus of the first interface, determine that the data transfer rate falls below a predetermined threshold and there is no command in the command queue, enter a pseudo-idle power state in which input and output (IO) drivers of the second interface is turned off but a power

domain for an on-chip internal memory of the storage controller is kept on, determine that there is a new command in the command queue from the host, and exit the pseudo-idle power state and enter the active power state.

(31) In one embodiment, the predetermined threshold of the data transfer rate may be 500 megabytes per second (MB/s).

(32) In one embodiment, the data transfer rate may be an average bus transfer rate calculated as a size of data transferred within a second.

(33) In one embodiment, the data transfer rate may be checked at a frequency of once every millisecond.

(34) In one embodiment, the processor may be further configured to exit the pseudo-idle power state and enter a sleep state when the data transfer rate has been zero for a first threshold of time and there is no new command in the command queue from the host. The first threshold of time may be in a range of 1 to 3 seconds.

(35) In one embodiment, in the pseudo idle power state, the processor may be further configured to turn off on-die termination (ODT) circuits of the second interface.

(36) In one embodiment, in the pseudo idle power state, the processor may be further configured to turn on clock gating and maintain a clock speed for the storage controller that is lower than in the active power state but higher than in any idle power states.

(37) In another exemplary embodiment, there is provided a method for managing power of a non-volatile storage system. The method may comprise handling data transfer requests from a host in an active power state, monitoring a command queue in a storage controller of the non-volatile storage system and a data transfer rate on a data bus of a first interface between the host and the non-volatile storage system, determining that the data transfer rate falls below a predetermined threshold and there is no command in the command queue, entering a pseudo-idle power state in which input and output (IO) drivers of a second interface coupling the storage controller and a non-volatile storage device of the non-volatile storage system is turned off but a power domain for an on-chip internal memory of the storage controller is kept on, determining that there is a new command in the command queue from the host, and exiting the pseudo-idle power state and entering the active power state.

(38) In one embodiment, the predetermined threshold of the data transfer rate may be 500 megabytes per second (MB/s).

(39) In one embodiment, the data transfer rate may be an average bus transfer rate calculated as a size of data transferred within a second.

(40) In one embodiment, the data transfer rate may be checked at a frequency of once every millisecond.

(41) In one embodiment, the on-chip internal memory of the storage controller may be a Static Random Access Memory (SRAM).

(42) In one embodiment, the method may further comprise: turning off on-die termination (ODT) circuits of the second interface in the pseudo idle power state.

(43) In one embodiment, the method may further comprise, in the pseudo idle power state, turning on clock gating and maintaining a clock speed for the storage controller that is lower than in the active power state but higher than in any idle power states.

(44) In yet another exemplary embodiment, there is provided a non-transitory machine-readable medium having computer instructions embedded thereon. The computer instructions, when executed by a hardware processor, may cause the hardware processor to perform: handling data transfer requests from a host at a non-volatile storage system in an active power state, monitoring a command queue in a storage controller of the non-volatile storage system and a data transfer rate on a data bus of a first interface between the host and the non-volatile storage system, determining that the data transfer rate falls below a predetermined threshold and there is no command in the command queue, entering a pseudo-idle power state in which input and output (IO) drivers of a

second interface coupling the storage controller and a non-volatile storage device of the non-volatile storage system is turned off but a power domain for an on-chip internal memory of the storage controller is kept on, determining that there is a new command in the command queue from the host, and exiting the pseudo-idle power state and entering the active power state.

(45) In one embodiment, the predetermined threshold of the data transfer rate may be 500 megabytes per second (MB/s).

(46) In one embodiment, the data transfer rate may be an average bus transfer rate calculated as a size of data transferred within a second, and checked at a frequency of once every millisecond.

(47) In one embodiment, the computer instructions, when executed by the hardware processor, may further cause the hardware processor to perform: exiting the pseudo-idle power state and entering a sleep state when the data transfer rate has been zero for a first threshold of time and there is no new command in the command queue from the host. The first threshold of time may be in a range of 1 to 3 seconds.

(48) In one embodiment, the computer instructions, when executed by the hardware processor, may further cause the hardware processor to perform: turning off on-die termination (ODT) circuits of the second interface in the pseudo idle power state.

(49) In one embodiment, the computer instructions, when executed by the hardware processor, may further cause the hardware processor to perform, in the pseudo idle power state, turning on clock gating and maintaining a clock speed for the storage controller that is lower than in the active power state but higher than in any idle power states.

(50) Any of the disclosed methods and operations may be implemented as computer-executable instructions (e.g., software code for the operations described herein) stored on one or more computer-readable storage media (e.g., non-transitory computer-readable media, such as one or more optical media discs, volatile memory components (such as DRAM or SRAM), or nonvolatile memory components (such as hard drives)) and executed on a device controller (e.g., firmware executed by ASIC). Any of the computer-executable instructions for implementing the disclosed techniques as well as any data created and used during implementation of the disclosed embodiments can be stored on one or more computer-readable media (e.g., non-transitory computer-readable media).

(51) While various aspects and embodiments have been disclosed herein, other aspects and embodiments will be apparent to those skilled in the art. The various aspects and embodiments disclosed herein are for purposes of illustration and are not intended to be limiting, with the true scope and spirit being indicated by the following claims.

Claims

1. A storage controller comprising a command queue and a processor; wherein the processor is configured to, determine a data transfer rate from a host and whether the command queue containing any command, wherein the processor is configured to, in response to the data transfer rate falling below a predetermined non-zero threshold and the command queue containing no command, enter a pseudo-idle power state by turning off input and output (IO) drivers of a memory interface coupling the storage controller and a non-volatile memory device and keeping on an internal memory of the storage controller; wherein the processor is configured to, in response to the data transfer rate reaching zero and staying zero for a first temporal threshold, enter a sleep state from the pseudo-idle power state, by putting the internal memory to sleep, turning off a first power domain of the storage controller, and maintaining a clock speed for the storage controller that is lower than in the pseudo-idle power state.

2. The storage controller of claim 1, wherein the processor is configured to: handle data transfer requests from the host in an active power state; or monitor the command queue and the data transfer rate; or determine whether the data transfer rate falls below the predetermined non-zero

threshold and whether the command queue contains no command.

3. The storage controller of claim 1, wherein the processor is configured to, in response to the command queue containing a new command from the host, exit the pseudo-idle power state and enter an active power state.

4. The storage controller of claim 3, wherein the processor is configured to determine whether the command queue contains the new command.

5. The storage controller of claim 1, wherein the data transfer rate is an average bus transfer rate calculated as a size of data transferred within a second.

6. The storage controller of claim 1, wherein the processor is configured to exit the pseudo-idle power state and enter a sleep state in response to the data transfer rate being zero for a first duration and the command queue containing no command.

7. The storage controller of claim 1, wherein in the pseudo-idle power state, the processor is configured to turn off on-die termination (ODT) circuits of the memory interface.

8. The storage controller of claim 1, wherein in the pseudo-idle power state, the processor is configured to turn on clock gating and maintain a clock speed for the storage controller that is lower than in an active power state.

9. The storage controller of claim 1, wherein the processor is configured to, in response to the data transfer rate staying zero for a second temporal threshold longer than the first temporal threshold, enter a deep sleep state from the sleep state, by turning off a second power domain of the storage controller.

10. A non-volatile storage system, comprising: the storage controller of claim 1; a host interface coupling the storage controller to the host; the non-volatile memory device; and the memory interface.
